



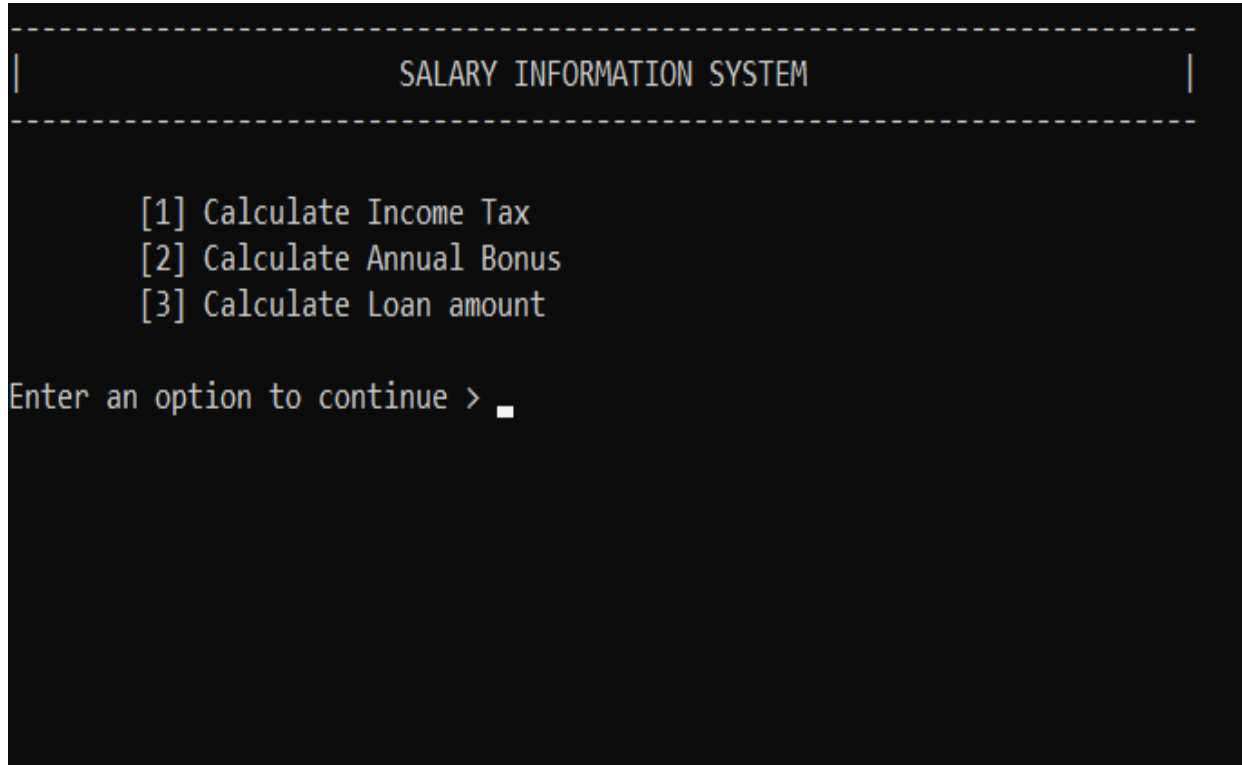
iCET Certified Developer

COURSEWORK

Module	Programming Fundamentals
Coursework Title	Salary Information System
Coursework Type	Practical Exercise

Salary Information System

This is a very simple application to calculate three different types of salary-related information: income tax, annual bonus, and loan amounts. When running this program, prompt the window to get user input for select option(Figure 1). After getting option number from user, the system should prompt the selected option.

A screenshot of a terminal window with a black background and light blue text. At the top, the title 'SALARY INFORMATION SYSTEM' is centered between two vertical bars. Below the title, three menu items are listed: '[1] Calculate Income Tax', '[2] Calculate Annual Bonus', and '[3] Calculate Loan amount'. At the bottom, a prompt 'Enter an option to continue >' is followed by a small white cursor block.

```
-----  
|                SALARY INFORMATION SYSTEM                |  
-----  
  
[1] Calculate Income Tax  
[2] Calculate Annual Bonus  
[3] Calculate Loan amount  
  
Enter an option to continue > █
```

Figure 1 – Salary Information System Home Page

01. Calculate Income Tax ([Demo](#))

The income tax that an employee has to pay based on their salary. The percentage of monthly Income Tax is given below.

- Income Tax is charged above Rs.100 000 and no tax is charged up to Rs.100 000.
- Rs.100 000 to the 1st Rs.41 667 (100 000 – 141 667) - 6%
- Rs.141 667 to the 2nd Rs.41 667 (141 667 – 183 333) - 12%
- Rs.183 333 to the 3rd Rs.41 667 (183 333 – 225 000) - 18%
- Rs. 225 000 to the 4th Rs.41 667 (225 000 – 266 667) - 24%
- Rs.266 667 to the 5th Rs.41 667 (266 667 – 308 333) - 30%
- More than Rs.308 333 - 36%

Monthly Income	Tax rate	Tax
100,000	NIL	
141,667	6%	2,500
183,333	12%	7,500
225,000	18%	15,000
266,667	24%	25,000
308,333	30%	37,500
350,000	36%	52,500
450,000	36%	88,500
550,000	36%	124,500
650,000	36%	160,500
750,000	36%	196,500
900,000	36%	250,500
1,000,000	36%	286,500

Several monthly incomes and the total Income Tax are given in the above table. When the user input Employee name and salary, system should print monthly Income tax(Figure 2).

Ex:- If an employee receives Rs.250 000 salary per month, his total amount of monthly Income Tax is Rs.21 000.

```

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|                               Calculate Income Tax                               |
-----
Input Employee name   - Rusiru
Input Employee salary - 250000

You have to pay Income Tax per month: 21000

```

Figure 2 – Calculate Income Tax

02. Calculate Annual Bonus ([Demo](#))

The annual bonus that an employee is eligible for based on their salary. The following table shows how bonus is calculated for Employees. When the user input Employee name and salary, system should print annual bonus (figure 3).

Monthly Income	Annual Bonus
Less than Rs.100 000	Rs.5000
Rs.100 000 - Rs.199 999	10%
Rs.200 000 - Rs.299 999	15%
Rs.300 000 - Rs.399 999	20%
More than Rs.400 000	35%

```

-----
|                               Calculate Annual Bonus                               |
-----
Input Employee name   - Rusiru
Input Employee salary - 250000
Annual bonus         - 37500.0

```

Figure 3 - Calculate Annual Bonus

03. Calculate Loan amount ([Demo](#))

First, the user should enter employee's name and monthly salary. Employee loan annual interest rate is 15%. The user should enter the number of years. The maximum limit that an employee can take a loan is to pay an installement of 60% of the monthly salary and employee salary should grater than Rs.50000 to get a loan (Figure 5). The maximum number of the year is 5. Finally , system should calculate the loan amount that can be obtained according to the employee's salary (Figure 4).

When the user input the employee salary and the number of year, we can calculate loan amount by using leasing formula. Remember, loan amount round of nearest 1000.

Lease Payment Calculation Formula

$$\text{Loan amount} = \text{Monthly installment} \times \frac{(1 - (1 / (1 + (r/12))^n))}{(r / 12)}$$

- **n** – number of months
- **r** - annual interest rate

```

-----
|                               Calculate Loan amount                               |
-----

Input Employee name   - Rusiru
Input Employee salary - 250000
Enter number of year : 3

You can get Loan Amount : 4327000

```

Figure 4 - Calculate Loan Amount

```

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|                               Calculate Loan amount                               |
-----

Input Employee name   - Shehan
Input Employee salary - 45000
    You can not get a loan because your salary less than Rs.50 000...

```

Figure 5 – Calculate Loan Amount Not Qualified Message

Guidelines

1. Demo Video Reference

- Click this link to refer demo video
- The demo video is provided to help you better understand the task requirements and clarify any doubts you may have.

2. Programming Requirements

- Use the Scanner class to obtain input from the command-line interface.
- Ensure your program runs correctly and reflects your individual understanding.

3. Submission Instructions

- Follow the instructions provided in the LMS.
- Submit your Java code without changing the file name on or before the coursework deadline.

4. Academic Integrity Policy

- This task must be completed individually.
- Do not use AI tools, code generators, or copy answers from any external sources or other students.
- Submissions must represent your original work and effort.

5. Support

- If you require any clarification, please feel free to contact your instructor before the deadline.